

Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional customer data to identify patterns in purchasing, spending, product performance, customer loyalty, subscription behavior, and discount usage.

The project follows an end-to-end data analytics workflow:

Data Preparation → MySQL Database → SQL Analysis → Power BI → DAX → Interactive Dashboard → Business Insights

The objective was not only to analyze the dataset but also to build an interactive business intelligence dashboard that allows users to explore customer and purchasing behavior from multiple perspectives.

2. Project Objectives

The main objectives of the project were to:

- Understand overall customer purchasing behavior.
- Analyze revenue and spending patterns across customer groups.
- Identify high-performing products and categories.
- Examine customer loyalty and repeat purchasing behavior.
- Analyze subscription behavior.
- Evaluate discount usage and its relationship with purchasing.
- Compare shipping and payment preferences.
- Identify patterns in customer purchase frequency.
- Build an interactive Power BI dashboard for business users.
- Derive actionable business recommendations from the analysis.

3. Dataset Overview

The dataset contains **3,900 customer purchase records and 18 columns**.

The dataset includes information covering:

Customer Information

- Customer ID
- Age
- Gender
- Location
- Subscription Status

Purchase Information

- Item Purchased

- Category
- Purchase Amount
- Season
- Size
- Color

Shopping Behavior

- Discount Applied
- Promo Code Used
- Previous Purchases
- Frequency of Purchases
- Review Rating
- Shipping Type
- Payment Method

Initially, the dataset contained **37 missing values in the Review Rating column**.

4. Data Preparation Using Python

Python and pandas were used for the initial exploration, cleaning, and transformation of the dataset.

4.1 Data Loading

The CSV dataset was loaded into Python using pandas.

Initial exploratory checks were performed using:

- `head()` to inspect the first records.
- `info()` to understand the dataset structure and data types.
- `describe()` to examine numerical statistics.
- `isnull().sum()` to identify missing values.

4.2 Missing Value Handling

The dataset contained missing values in the Review Rating column.

Instead of replacing all missing ratings with one overall value, the missing ratings were filled using the **median review rating within each product category**.

This approach preserved differences in rating patterns between categories and reduced the influence of extreme values.

4.3 Column Standardization

The original column names were converted into a consistent lowercase and snake-case format.

For example:

Purchase Amount (USD)

was standardized to:

`purchase_amount`

This made the columns easier to work with in Python, SQL, and Power BI.

4.4 Feature Engineering

Two additional columns were created during data preparation.

Age Group

Customers were divided into four age groups using quartile-based segmentation:

- Young Adult
- Adult
- Middle Aged
- Senior

This created a more useful categorical variable for demographic analysis.

Purchase Frequency Days

The original purchase-frequency categories were converted into approximate numbers of days.

Examples:

- Weekly → 7 days
- Fortnightly → 14 days
- Monthly → 30 days
- Quarterly → 90 days
- Annually → 365 days

This transformation allowed purchase frequency to be analyzed numerically.

4.5 Data Consistency Check

The relationship between `discount_applied` and `promo_code_used` was checked.

The values were found to be consistent, meaning the two columns were effectively providing redundant information.

Therefore, the promo_code_used column was removed from the cleaned dataset.

5. MySQL Database Integration

After the Python-based preparation stage, the cleaned dataset was loaded into **MySQL**.

A MySQL schema named:

customer_behaviour

was created for the project.

Python was connected to MySQL using:

- PyMySQL
- SQLAlchemy

The cleaned DataFrame was then loaded into a MySQL table named:

customer

This created the database layer used for the SQL analysis.

Workflow

CSV Dataset

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Python / pandas

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Data Cleaning & Feature Engineering

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MySQL — customer_behaviour schema

↓

customer table

6. SQL Data Analysis

SQL was used to answer a series of business questions from the cleaned customer dataset.

The analysis was performed directly in MySQL.

6.1 High-Spending Customers Using Discounts

Customers who used discounts and whose purchase amount was greater than or equal to the overall average purchase amount were identified.

This helped identify customers who continued to make relatively high-value purchases while receiving discounts.

6.2 Top Products by Average Rating

The average review rating was calculated for each purchased item.

The products were sorted by average rating in descending order, and the **top 5 products** were identified.

This analysis helps identify products receiving the strongest customer feedback.

6.3 Average Purchase Amount by Shipping Type

Average purchase amounts were compared between:

- Standard shipping
- Express shipping

This provided insight into whether customers using different shipping options showed differences in spending.

6.4 Subscriber vs Non-Subscriber Analysis

Customers were grouped according to subscription status.

For each group, the analysis calculated:

- Number of customers
- Average purchase amount
- Total revenue

This allowed subscriber and non-subscriber purchasing behavior to be compared.

6.5 Products with the Highest Discount Rate

The percentage of purchases made with a discount was calculated for each product.

The products were then ranked by discount rate, with the top 5 identified.

This analysis helps identify products that are more frequently purchased with discounts.

6.6 Customer Segmentation

Customers were segmented according to their previous purchase history.

The following segments were created:

- **New:** 1 previous purchase
- **Returning:** 2–10 previous purchases
- **Loyal:** More than 10 previous purchases

The number of customers in each segment was then calculated.

This provides a simple view of customer loyalty based on purchasing history.

6.7 Top 3 Products within Each Category

A window function using ROW_NUMBER() was used to rank products within each category based on order count.

The top 3 products from each category were identified.

This provided a category-level view of the most frequently purchased products.

6.8 Repeat Buyers and Subscription Status

Customers with more than 5 previous purchases were treated as repeat buyers.

These customers were grouped by subscription status to examine the relationship between repeat purchasing and subscriptions.

6.9 Revenue by Age Group

Total purchase revenue was calculated for each age group.

The groups were then ranked by their revenue contribution.

This helped identify which customer age groups contribute the most revenue.

7. Power BI Integration

After completing the SQL analysis, the MySQL database was connected to **Power BI**.

The customer table from the customer_behaviour schema was imported into Power BI.

This allowed the cleaned and SQL-ready dataset to be used for interactive visualization and dashboard development.

8. DAX Measures

Several DAX measures were created to support the Power BI dashboard.

The core measures included:

Number of Customers

No. of customers =
`COUNT('customer_behaviour customer'[customer_id])`

This measure calculates the total number of customer records.

Average Purchase Amount

Avg purchase amount =
`AVERAGE('customer_behaviour customer'[purchase_amount])`

This calculates the average amount spent per purchase.

Average Review Rating

Avg review rating =
`AVERAGE('customer_behaviour customer'[review_rating])`

This calculates the average customer review rating.

Additional measures were created during dashboard development for revenue, discount behavior, subscription rate, purchase frequency, and other analytical requirements.

9. Power BI Dashboard Development

The final Power BI dashboard was designed as a **four-page interactive analytical report**.

The pages were designed to answer different business questions rather than repeating the same type of visualization.

10. Page 1: Executive Overview

The first page provides a high-level view of the business.

It focuses on overall customer and purchasing performance.

Key metrics include:

- Number of Customers
- Total Revenue
- Average Purchase Amount
- Average Review Rating

- Subscription Rate

The page also provides high-level comparisons across customer demographics and purchasing characteristics.

This page acts as the executive summary of the dashboard.

11. Page 2: Customer & Loyalty Analysis

The second page focuses on understanding customers and their purchasing behavior.

It analyzes areas such as:

- Customer segments
- New vs returning vs loyal customers
- Subscription behavior
- Customer demographics
- Purchase behavior across customer groups

This page helps answer:

Who are the customers, and how does their purchasing behavior differ across segments?

12. Page 3: Product & Category Performance

The third page focuses on product and category-level performance.

Top Products by Revenue

A ranked bar chart was created to identify the highest-revenue products.

An interactive **Top 5 / Top 10 toggle** was implemented using:

- Two overlapping visuals
- Bookmarks
- Selection Pane
- Buttons

The button works as a toggle, allowing the user to switch between the two views.

Revenue Contribution by Category

A treemap was used to visualize the contribution of different product categories to total revenue.

Discount Rate by Category

A bar chart compares discount rates across categories.

Average Customer Rating by Category

A column chart compares average review ratings across categories.

This page combines product ranking, category performance, discount behavior, and customer feedback.

13. Page 4: Purchase Behavior & Revenue Analysis

The fourth page focuses on deeper purchasing patterns and revenue analysis.

Previous Purchases vs Purchase Amount

A scatter plot was used to explore the relationship between previous purchasing behavior and purchase amount.

The visualization uses:

- Average previous purchases on the X-axis
- Purchase amount on the Y-axis
- Category as the legend

This provides an exploratory view of purchasing patterns across categories.

Revenue Decomposition

A **Decomposition Tree** was used to allow interactive exploration of total revenue.

Revenue can be broken down through dimensions such as:

- Category
- Age Group
- Gender
- Subscription Status
- Payment Method

This allows users to interactively explore different contributors to revenue.

Revenue by Payment Method

A donut chart was used to compare revenue generated through different payment methods.

Purchase Frequency by Category

A column chart compares purchase frequency across categories.

The dashboard also includes an **Avg Days Between Purchases** KPI to make the frequency metric easier to interpret.

14. Interactive Dashboard Features

The dashboard includes several interactive features to improve usability.

Slicers

Common filters were included across the dashboard for:

- Age Group
- Gender
- Category
- Subscription Status

These allow users to dynamically filter the dashboard.

Page Navigation

Navigation buttons allow users to move between the four analytical pages.

Bookmark-Based Top 5 / Top 10 Toggle

The Product Performance page includes a custom interactive toggle.

The user sees a single button representing the alternative view:

Top 5 ↔ Top 10

Bookmarks control the visibility of the corresponding visuals and buttons.

This creates a cleaner interactive experience without displaying both charts simultaneously.

Decomposition Tree

The Decomposition Tree provides an additional interactive method for exploring revenue drivers.

15. Key Analytical Areas

The completed project provides analysis across several major business areas:

Customer Behavior

- Customer segments
- Repeat purchasing
- Purchase frequency
- Demographic patterns

Product Performance

- Top products
- Category revenue
- Product ratings

- Discount dependency

Revenue Analysis

- Revenue by age group
- Revenue by category
- Revenue by payment method
- Subscriber vs non-subscriber revenue

Marketing & Discounts

- Discount usage
- Discount rates by product/category
- High-spending customers using discounts

Customer Loyalty

- New customers
- Returning customers
- Loyal customers
- Repeat buyers
- Subscription behavior

16. Business Recommendations

Based on the analytical areas covered in the project, the following recommendations can be considered:

Strengthen Customer Loyalty

Develop targeted loyalty initiatives for returning customers and encourage repeat buyers to become long-term loyal customers.

Optimize Subscription Strategy

Compare subscriber and non-subscriber spending and revenue patterns to identify opportunities for improving subscription adoption.

Review Discount Strategy

Identify products with particularly high discount rates and evaluate whether discounts are contributing to meaningful purchasing behavior.

Promote High-Performing Products

Use high-revenue, highly rated, and frequently purchased products in promotional campaigns and product positioning.

Target High-Value Customer Groups

Use demographic and purchasing patterns to focus marketing efforts on customer groups contributing strongly to revenue.

Improve Data-Driven Decision Making

Use the Power BI dashboard as an interactive reporting layer so business users can filter, compare, and investigate customer behavior without manually running SQL queries.

17. Tools & Technologies

Area	Technology
Data Processing	Python
Data Manipulation	pandas
Database	MySQL
Python–Database Connection	PyMySQL, SQLAlchemy
Querying & Analysis	SQL
Visualization	Power BI
Calculations	DAX
Dashboard Interactivity	Power BI Bookmarks, Buttons, Slicers
Advanced Visuals	Decomposition Tree, Scatter Plot, Treemap

18. Conclusion

The Customer Shopping Behavior Analysis project demonstrates an end-to-end data analytics workflow, starting from raw transactional data and progressing through data preparation, database management, SQL analysis, DAX calculations, visualization, and business intelligence reporting.

The final four-page Power BI dashboard provides an interactive way to analyze customer behavior, product performance, purchasing patterns, revenue contribution, discounts, subscriptions, and customer loyalty.

The project combines technical data analytics skills with business-oriented interpretation, making it suitable as a portfolio project demonstrating practical experience with **Python, SQL, MySQL, Power BI, and DAX**.